SC2014 : Sensors, Interfacing & Digital Control

Part 2 : Tutorial 4 – Transient Analysis & Stability

Q1 : The transfer function of an open-loop chemical plant, G(z), is given here. A controller function, C(z), is also given. Design a closed-loop unity gain negative feedback system for this plant and the controller. Find the Controller Gain. K, for the system to have a damping ratio of 0.3, a natural undamped frequency of 70 rad/s and a sampling rate of 20 samples/sec.

$$G(z) = \frac{1}{z^2 + 0.686z + 0.002} \qquad C(z) = \frac{K}{200}$$

Q2 : The system characteristic equation of the mechanical system is given as :

$$P(z) = z^2 - 1.321z + 0.497$$

Find the damping ratio, ζ, and undamped natural frequency, ω_n for this system when the sampling period is 0.2s.

Q3 : Consider the following open-loop transfer function in the z-domain

$$G(z) = \frac{1}{(z-1)(z-0.3)}$$

With *T = 0.1s*, design the closed-loop unity gain negative feedback system that includes a proportional controller. Find the proportional gain, ***K***, to achieve the followings:

i) A damped natural frequency of 5 rad/s.
ii) A settling time of 2s.

You may assume the following relationship between the damped natural frequency (ω_d) and undamped natural frequency (ω_n) :

$$\omega_d = \omega_n\sqrt{1-\delta^2}$$

For Question 1 to 3. You may assume the standard characteristic equation for a second order system is (either version can be used) :

$$z^2 - 2\cos\left(\omega_n T\sqrt{1-\delta^2}\right)e^{-\delta\omega_n T}z + e^{-2\delta\omega_n T}$$

$$z^2 - 2\cos(\omega_d T)e^{-\delta\omega_n T}z + e^{-2\delta\omega_n T}$$

Q4 : Examine the stability of the characteristic equation given by :

$$P(z) = (z-1)(z-0.8)(z+0.5)$$

Plot the poles on the z-plane and comments on the positions of the poles and how they affect the stability of a system.

Apply Jury's stability test and comment on the stability of the system.

Q5 : Consider the system described by :

$$y(k) + 2.1y(k-1) + 0.4y(k-2) = x(k) + x(k-1)$$

where $y(k)$, $x(k)$ are output/input of the system. Determine the stability using Jury's stability test.